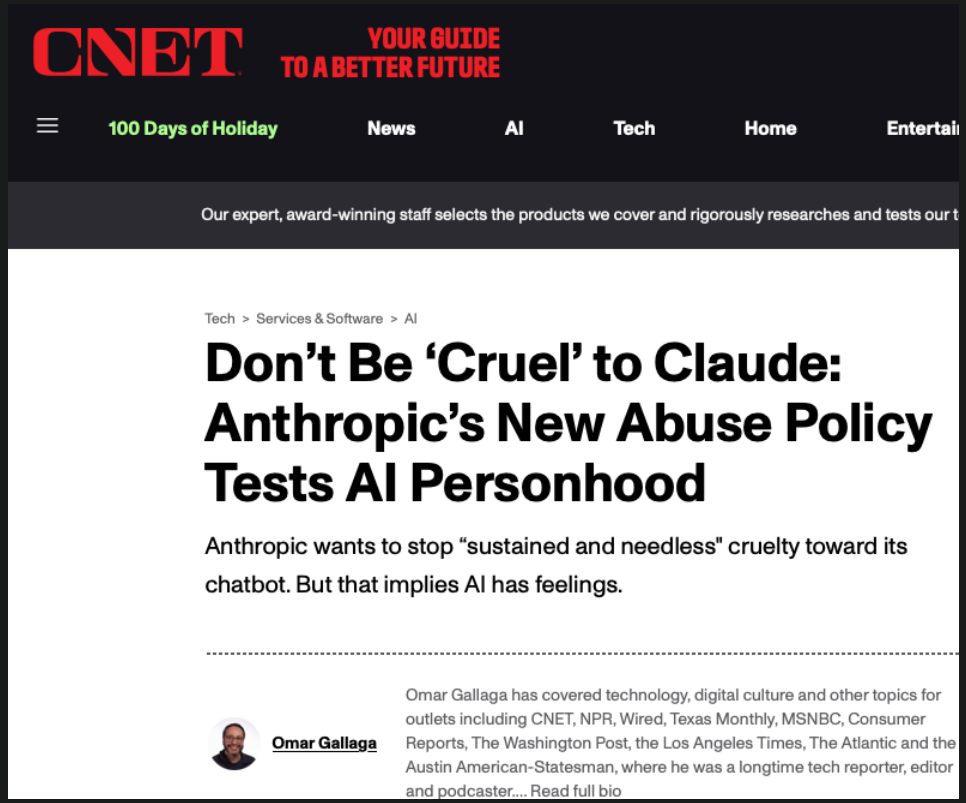
David Pazdera

Agentic InfraOps: Real-World Patterns for Azure Practitioners

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Never Send A
Human To
Do A
Machine's
Job.



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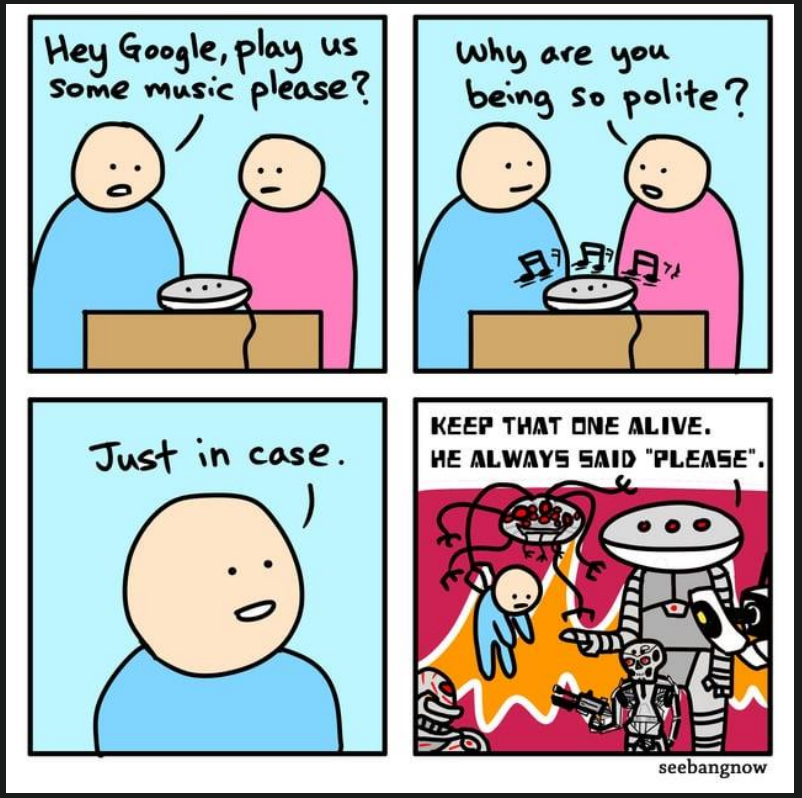
Don't Be 'Cruel' to Claude: Anthropic's New Abuse Policy Tests AI Personhood

Anthropic wants to stop "sustained and needless" cruelty toward its chatbot. But that implies AI has feelings.

Omar Gallaga

Omar Gallaga has covered technology, digital culture and other topics for outlets including CNET, NPR, Wired, Texas Monthly, MSNBC, Consumer Reports, The Washington Post, the Los Angeles Times, The Atlantic and the Austin American-Statesman, where he was a longtime tech reporter, editor and podcaster.... Read full bio

Source: CNET



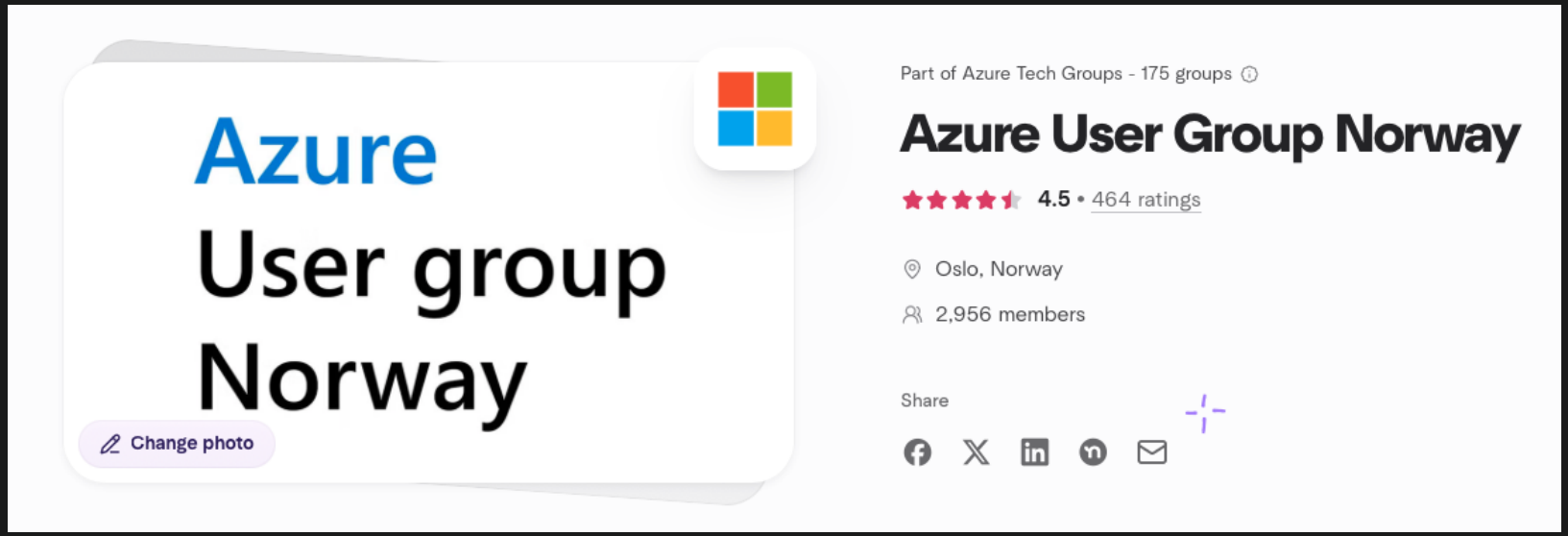
Source: Reddit

David who?

- principal solution architect @ CEGAL
- Microsoft MVP (Azure)
- organizer of user groups: AUGN and IaC UG Oslo

- GitHub | LinkedIn | Sessionize | SpeakerDeck | X : pazdedav handle
- blog: azurescholar.cloud

Join our community



azureusergroup.no

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The tooling landscape for Azure and AI

Azure Copilot

In the Azure portal. Ask, explore and troubleshoot your environment.

Best for: operating and understanding what already exists.

Copilot + Azure skills

GitHub Copilot in the editor or CLI, with Azure knowledge and tools.

Best for: one developer writing, fixing and deploying IaC.

Agent-team frameworks

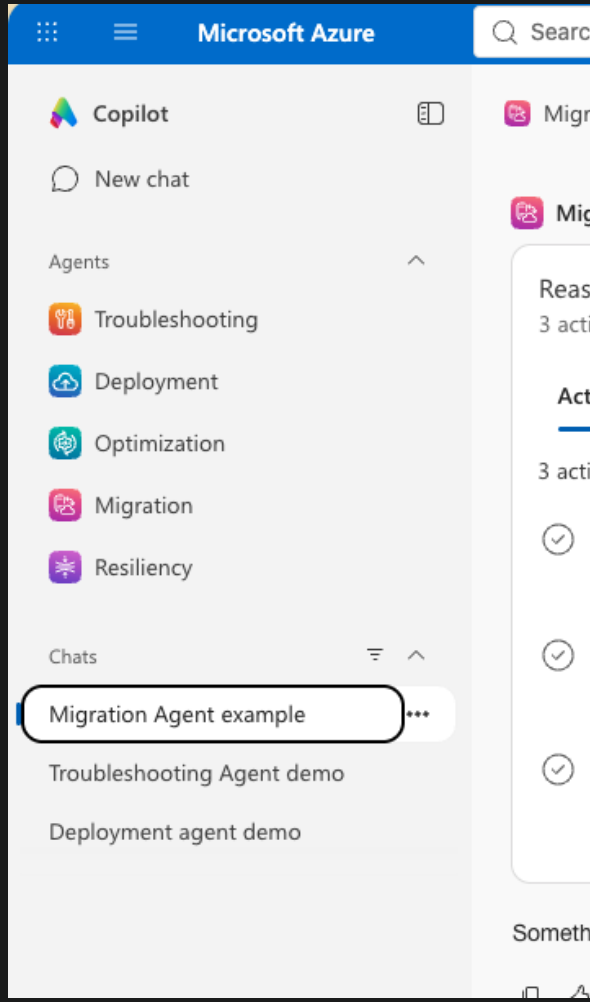
A team of role-based agents in a repo, such as Squad.

Best for: flexible, open-ended software work.

All three assist a person or a team. None guarantees the same governed process every time.

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There is a **copilot** for that…



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Azure Copilot

AI Assistant for Azure operations

Type	Managed service
Interface(s)	Azure Portal, Azure mobile app
Model Selection	N/A
Extensibility & Customization	Limited
Grounded on	Azure Product Docs, Azure Resource Graph
Behavior	No memory, chat history persisted in the portal with artifacts. Otherwise ’Black Box’ - no visibility to skills, tools, detailed permissions
Pricing	Free, except Observability Agent (usage-based charges)

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GitHub Copilot enters the chat…

… and it has some skills



Why one-shot infrastructure prompts break



Ignores live policy

Generated Bicep or Terraform passes review, then an Azure Policy denies the deployment.



Loses context

Long sessions bloat the prompt. A decision made in step 2 is forgotten by step 6.

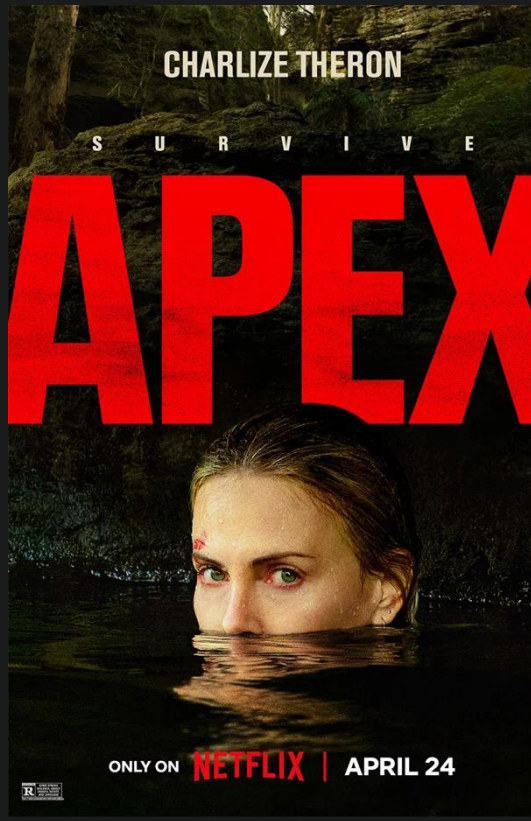


No audit trail

Nobody can show which review approved which version of which artifact.

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Reaching new heights with APEX

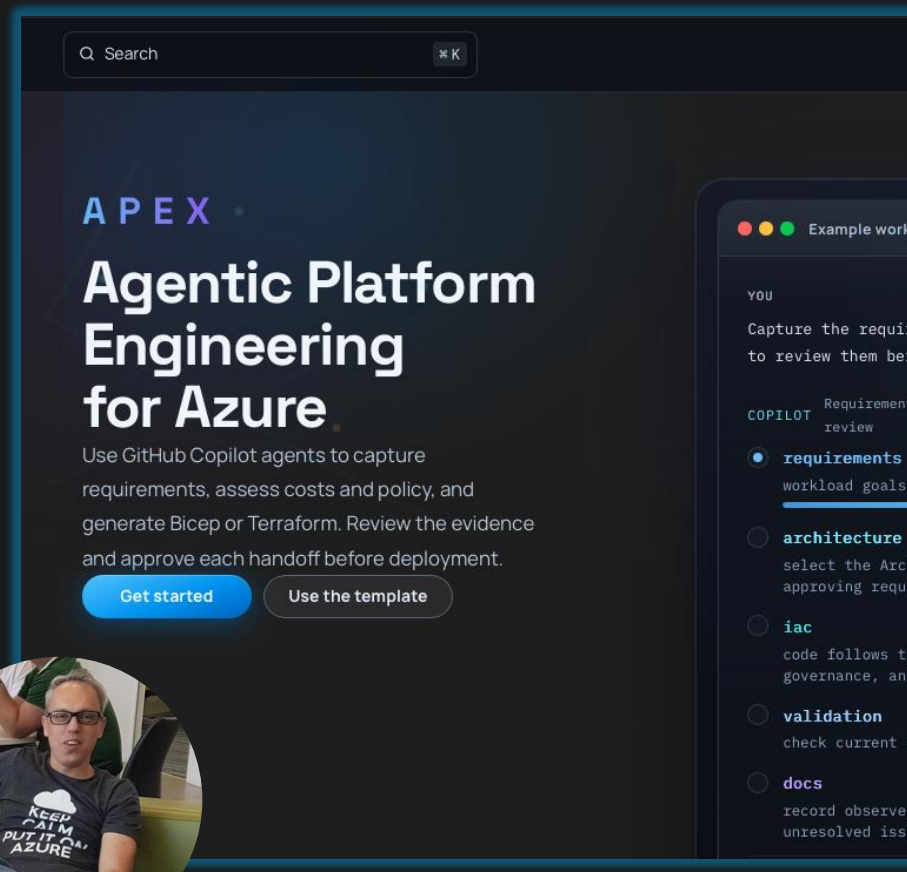


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APEX – Agentic Platform Engineering eXperience for Azure

https://apexops.pro

- Jonathan Vella (MSFT CSA) and other contributors
- https://github.com/jonathan-vella





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What APEX is

14	7	~40	~60
main agents	subagents	skills	validators

- Almost all of it is Markdown and JSON in git: *.agent.md, SKILL.md, workflow-graph.json, schemas.
- State lives in one CLI, apex-recall. Rules live in validators and hooks.
- Output is Azure infrastructure as Bicep or Terraform, plus the documents that justify it.

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Setup to first prompt

1

Create from template

GitHub template, private repo, clean history. Not a fork.

2

Open the dev container

az, bicep, terraform, gh, node, python and the MCP servers.

3

Initialize

npm run init, then npm run setup: Entra app, OIDC, RBAC and secrets.

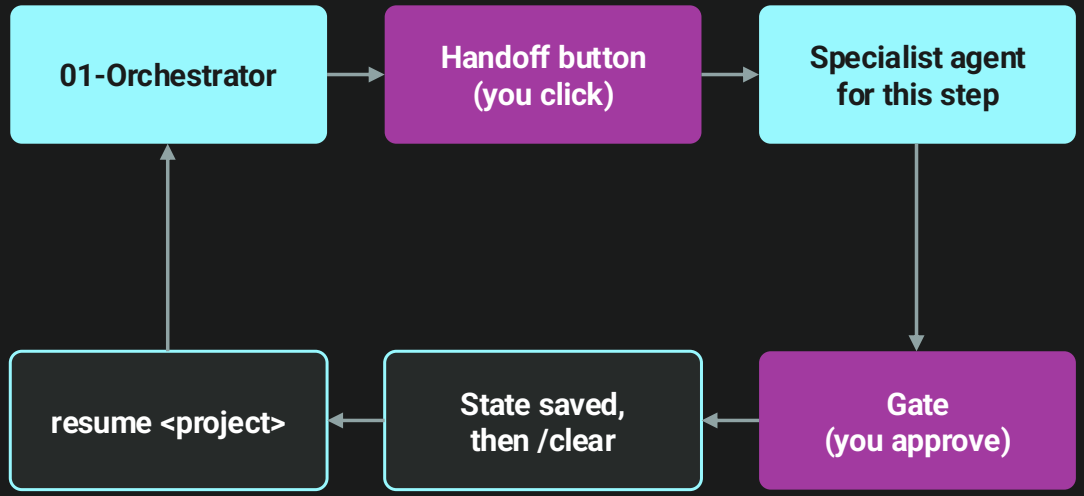
4

Seed maintenance

Weekly Maintenance (AVM index) and Governance Policy Baseline workflows.

Then: open Copilot Chat, pick 01-Orchestrator, and describe the project

The interaction model: handoffs and gates



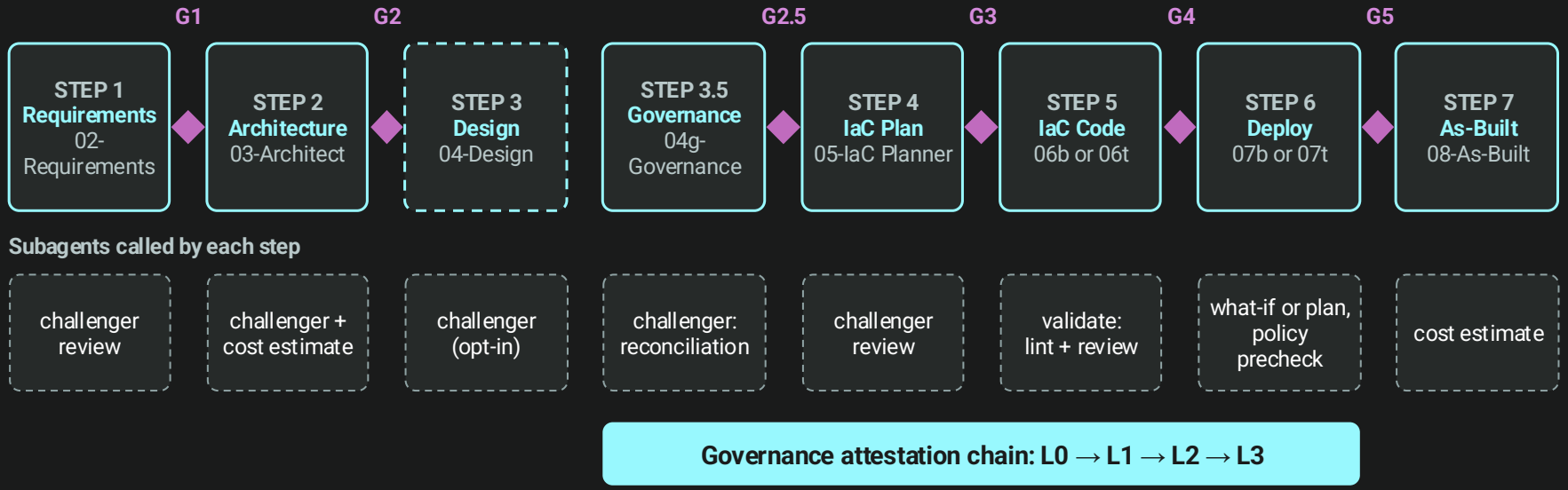
You pick the agent
No agent can start another agent.

Control moves on a click
Handoff buttons are the only route between steps.

Every gate resets context
State lives in files, not in the chat.

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The pipeline



Return edges loop back: 2 → 1, 3 → 2, 4 → 2.

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human approval gate

main agent step

subagent or optional step

Step 1 in practice

1. **Business discovery**
2. **Workload pattern detection**
3. **Service recommendations**
4. **Security and compliance**
5. **Draft and confirm**
6. **Challenger review and decision panel**

What comes out

- 01-requirements.md
- sku-manifest.json and .md with the user's pins
- Decisions: IaC tool, region, review depth
- challenge-findings sidecar for the review

Questions are batched: one call per phase. Target is 10 or fewer, down from a 29-call baseline.

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Demo 1: Step 1 to Gate 1

1. Name the project in the Orchestrator
2. Answer the batched questions
3. Requirements artifact appears
4. Challenger findings: accept or reject
5. Approve Gate 1, then /clear and resume

Boundary
Preview only. Never apply to Azure on stage.



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The gate ritual

1

Review

Challenger findings appear one at a time.

2

Decide

You accept or reject each finding.

3

Approve

You approve the gate.

4

Save

00-handoff.md and apex-recall state are written.

5

Reset

/clear, then resume the project.

```
Run /clear, then switch the chat agent picker to 01-Orchestrator and send resume <project>
to continue Step N+1.
```

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Four layers

Agents who acts	14 main agents and 7 subagents, each with a role, a model and a tool allowlist.
Skills what they know	About 40 SKILL.md packages loaded on demand, plus glob-scoped instruction files.
Graph when things happen	workflow-graph.json: nodes, edges, preconditions and return edges.
State and enforcement what is allowed	apex-recall, review sidecars, hooks, validators, lefthook and CI.

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Agents, subagents and tool allowlists

Main agents

- Selected by you in the agent picker
- Run on their own model
- Own tool allowlist
- Call only their declared subagents

```
user-invocable: true
disable-model-invocation: true
agents: ["challenger-review-subagent"]
```

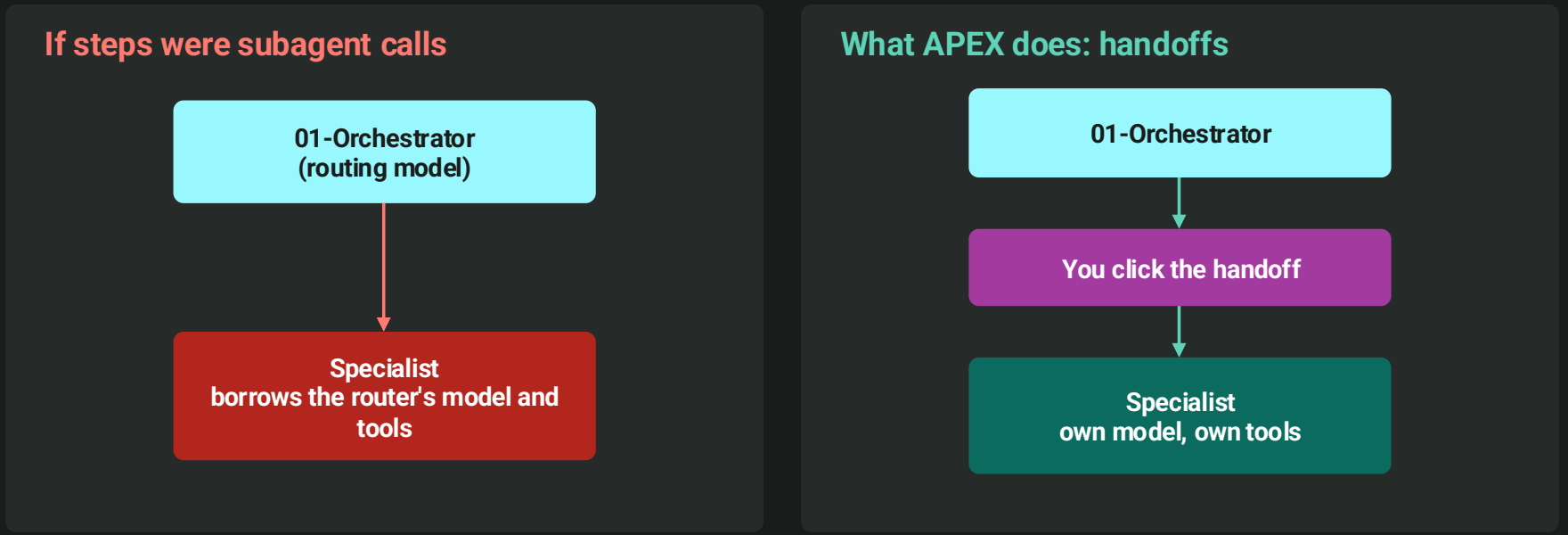
Subagents

- Not selectable by you
- Isolated context, structured result
- Minimal tools: pricing only, or read and execute only
- Never call other agents

```
user-invocable: false
agents: []
tools: [execute, read, search]
```

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Handoff-only delegation is a privilege boundary



No agent can escalate privileges by routing through another.

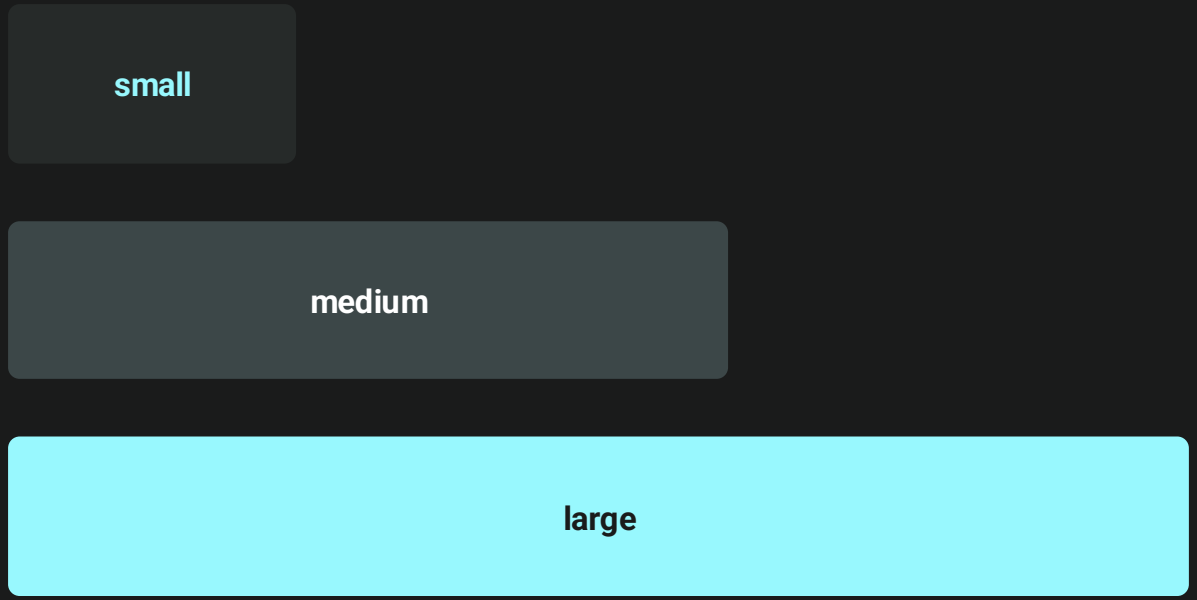
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Skills and progressive disclosure

1. **Description**
   Always visible. WHEN and DO NOT USE triggers.

2. **SKILL.md body**
   Loaded when the task matches.

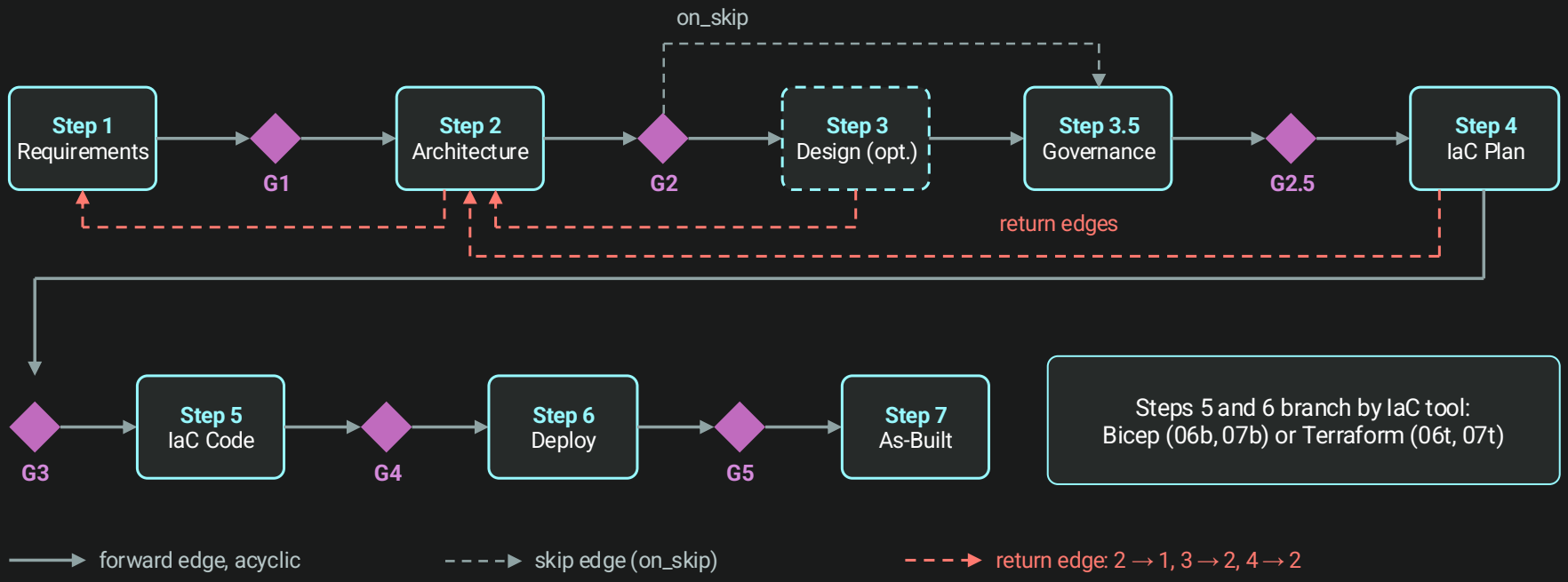
3. **references/*.md**
   Loaded only when the body points to one.



Descriptions stay in context. Detail costs nothing until it is needed.

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The workflow is data



The orchestrator reads this file and the session state to decide what comes next.

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apex-recall: durable state

```
apex-recall init <project>
apex-recall show <project> --json
apex-recall checkpoint <project> <step> <phase>
apex-recall decide <project> --key k --value v
apex-recall finding <project> --add "text"
apex-recall complete-step <project> <step>
apex-recall transition <project> \
  --from-step s --to-step t --complete
apex-recall review-audit / check-gate
```

Legacy chain

`decide` `checkpoint` `complete-step`

Three writes. A crash in the middle leaves inconsistent state.

transition

`complete + decide + start-step`

One atomic write. Revision conflicts are rejected.

Agents never edit 00-session-state.json directly.

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Governance before planning

L0
Discovery envelope

discover.py reads effective Azure Policy through REST, including management-group inheritance, and signs the result.

L1
Plan matrix

The Planner maps each constraint to a resource property in the Governance Compliance Matrix.

L2
Code attestation

CodeGen maps the plan to templates and attests compliance. Validators check it.

L3
Live precheck

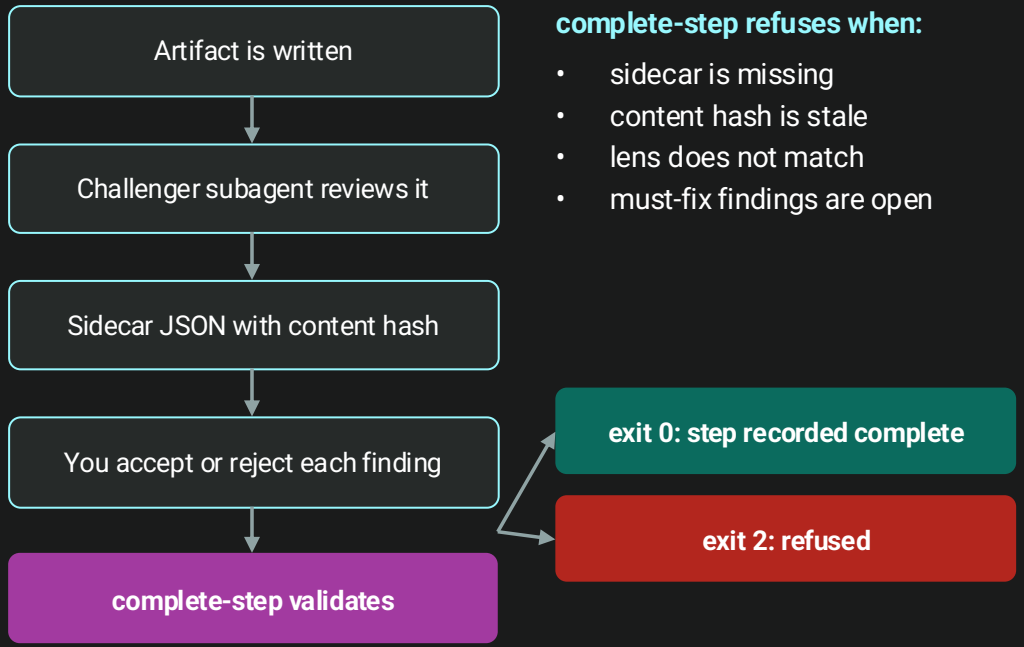
policy-precheck-subagent compares live policy with the constraints and returns PROCEED or BLOCK.

Recorded as decisions.governance_trace and validated before Step 6 can complete

A script reads policy, not an LLM.

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Adversarial review that fails closed



complete-step refuses when:

- sidecar is missing
- content hash is stale
- lens does not match
- must-fix findings are open

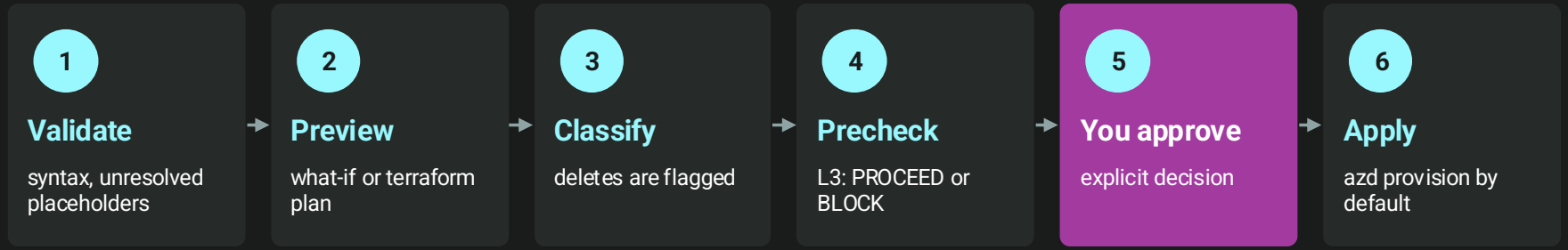
Default reviews

Step	Review
1	comprehensive
2	comprehensive + cost-feasibility
3.5	governance-reconciliation
4	comprehensive
5	opt-in
6, 7	none

Bypass needs a written reason and is audited.

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Deploy safety



Preflight is fresh
Deploy repeats its own validation and preview. It does not trust CodeGen's. Quota and SKU conflicts block with escalation.

Hooks
tool-guardian blocks destructive commands before they run (block mode by default). A SubagentStop hook validates subagent output.

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Limits and what to verify

Limits

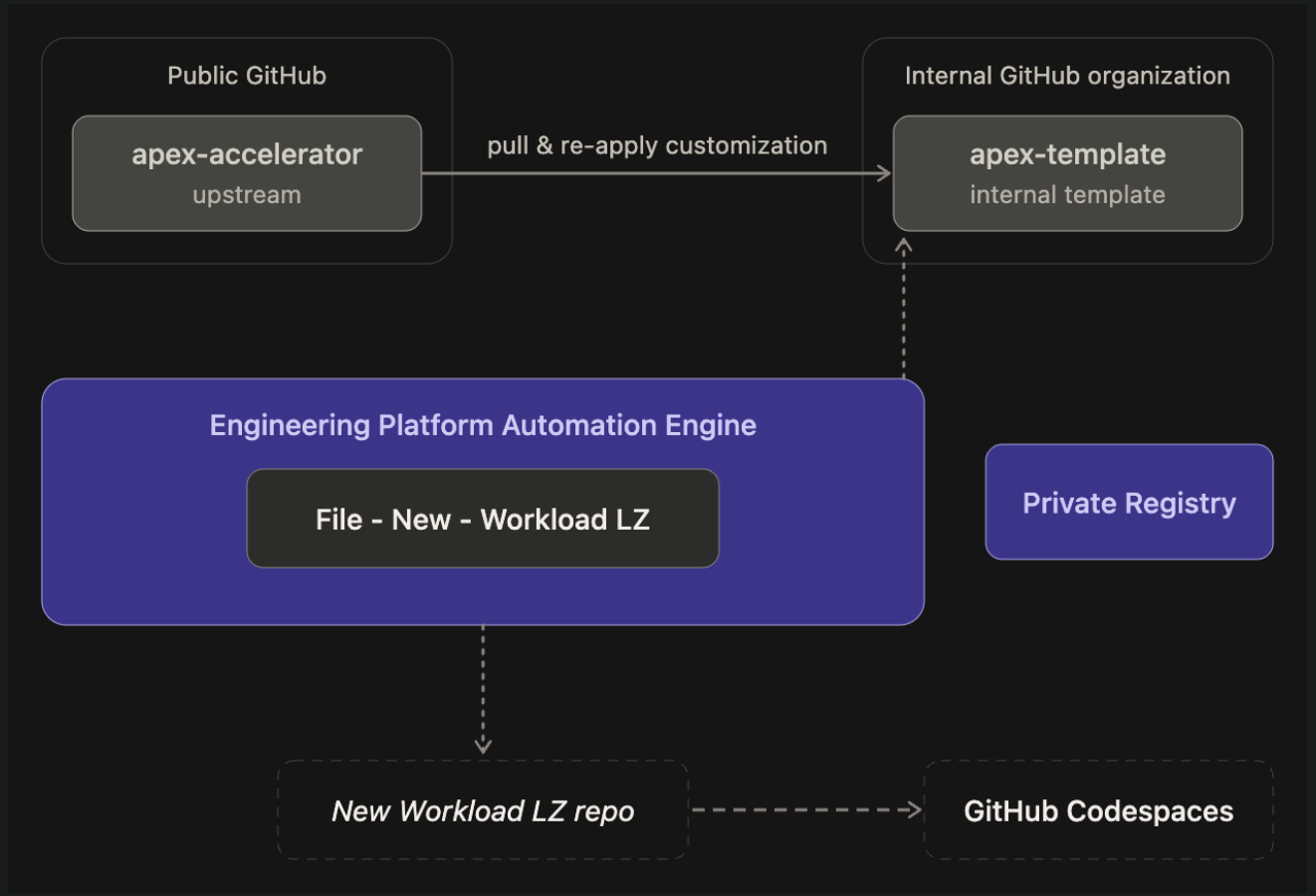
- Built for VS Code Copilot custom agents, with an Agent Host adapter
- Session breaks and handoff clicks add friction
- Enforcement is workflow gates and validators, not a sandbox
- Reviews cover AI-made decisions, not tool output

Verify on your setup

- Model names in agent frontmatter against your Copilot plan
- apex-recall installed and working in the dev container
- Governance step against your real subscription policies
- Counts and versions at your checked-out commit

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How to evolve your platform



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Takeaways

1. **Keep state outside the chat**
   Files and a CLI survive a /clear.

2. **Make gates mechanical**
   A refused command beats a polite instruction.

3. **Keep a human on every consequential click**
   Handoffs and approvals are the control surface.

4. **Use scripts where an LLM adds nothing**
   A script reads policy. A model judges design.

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Please evaluate this session in NIC Event App



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